Directed Energy Deposition JBK-75 Representative Process Samples and Component Builds Statement of Work (SOW)

NASA's Marshall Space Flight Center (MSFC) requires a large-scale manufacturing technology demonstration using thin-wall Laser Powder Directed Energy Deposition (DED). This effort aims to complete fabrication of material peroprty testing and characterization samples using JBK-75, NASA HR-1 and specialty alloys per NASA and industry specifications and representative process witness samples. These samples are required to be identical to part fabrication for metallography and mechanical and thermophysical testing.

The SOW is as follows:

Task 1

1. The vendor shall fabricate a series of witness test specimens to evaluate the processing of JBK-75 and NASA HR-1 according to NASA specification using laser powder directed energy deposition (LP-DED). The witness coupon specimens shall include round bars and harvest boxes, or otherwise specified by NASA, and be fabricated using identical machine setup, parameters, and process setup as the RS25 nozzle demonstrator.
2. The vendor shall fabricate and provide engineering support for the LP-DED to complete all build samples with an approximate wall thickness of 1 mm and integral channels, with representative machine and parameters as the full-scale RS25 nozzle demonstrator.
3. The vendor shall prepare build plates to support sample builds.
4. The vendor shall review final models and refine the build approach for the JBK-75 thin-wall RS-25 nozzle demonstrator.
5. The vendor shall provide design inputs and develop detailed toolpath strategies for confirmation by NASA.
6. Vendor shall demonstrate LP-DED using a large scale machine platform for capability to print the aft end of a RS-25 nozzle with a combined height of greater than 90” height and greater than 80” diameter.
7. Vendor shall provide capability within laser system for 1 mm wall thickness at diameters stated in Task 1, requirement #6.
8. The material shall be JBK-75 Fe-Ni and NSA HR-1 Fe-Ni alloys per NASA requirements. The use of virgin powder, 1x used and 2x used is acceptable.
9. The print shall be performed within 75 consecutive days, or approximately 1405 hours of machine time shall include the following:
    a. Engineering Support for a dedicated 557 laser system (includes Tool Code generation, part design input, laser scan inspections, reporting, etc.)
    b. Provide operator for machine setup and deposition.
    c. Provide consumables, such as Argon nozzles, filters, and all utilities required for operation.
    d. Provide a build volume at a minimum of 60” dia by 72” for component builds.
    e. Provide an inert gas chamber as part of the build to minimize oxidation. Provide all argon and consumables.

f. Provide tool code required to complete large format samples.
g. Provide all maintenance on the machine during the period of builds.
h. Provide operators necessary to operate the machine.
i. Shall include other required direct costs such as consumables and necessary replacement components (inert gas, build-nozzle components, filters, etc).
j. All engineering support should be included as part of the continuous-build machine capacity. This includes all tool path generation, inspections, trouble shooting of build failures, reporting, etc.
k. Shall include removal of parts from the build plate as required.

10. The vendor shall complete the following sample builds:

Part Number		Part Description	Quantity	Material
AR nozzle	RCA303007-3	MTD 20, SAMPLE, REPAIR, TRANSITION POINT	7	JBK-75
	RCA303009-3	MTD 22, SAMPLE, REPAIR, CONE WELD	8	JBK-75
	Age Test Sample	1/8"x.5" x 20" 1 x 1070 W Samples	1	JBK-75
	Age Test Sample	1/8"x.5" x 20" 1 x 350 W Samples	1	JBK-75
AR MCC	FWD LUGS	JBK-75, LP-DED, 2620W (6" x 2.5" x 0.6")	2	JBK-75
	AFT Lugs	JBK-75, LP-DED, 2620W (6" x 1.8" x 0.6")	2	JBK-75
	Base Ring	JBK-75, LP-DED, 2620W (6" x 1" x 1")	4	JBK-75
AR Materials	Material Characterization	Hollow Samples built with 350W build parameters	24	JBK-75
	Material Characterization	2" x 1" x 8" built with 1070W build parameters	2	JBK-75
MSFC	Material Characterization	0.5" dia x 4" length	100	JBK-75
	Material Characterization	0.5" dia x 4" length; Build Interruption halfway thorugh and pause	60	JBK-75
	Material Characterization	Thin Channel Plates, 8" length and 3" height (I-beam like)	6	JBK-75
	Material Characterization	12" dia channel builds ~4" height	4	JBK-75
	Material Characterization	Thin Channel Plates, 8" length and 3" height (I-beam like)	6	NASA HR-1
	Material Characterization	12" dia channel builds ~4" height	4	NASA HR-1

11. Part should be run on the same machine consecutively with acceptable breaks as needed for expected or unexpected shutdowns or pauses.
12. NASA shall provide drawings, models, and dimensions as required.
13. NASA will provide all heat treatments and characterization and inspections as necessary.
14. NASA shall provide DED-sized powder, specifically JBK-75, for the builds.
15. The witness coupons shall be removed from the build plate, unless otherwise specified.
16. The witness coupons shall be delivered to NASA in the as-built condition without any heat treatments.
17. The vendor shall complete a final inspection of the RS-25 nozzle aft end. The inspection

shall be completed using laser scanning and compared to the build CAD geometry, the interim scan data or both.

18. A report shall be provided of the scanning data from the RS25 aft end nozzle.
19. Purchase JBK-75 powder according to NASA specifications supporting task 1 not to exceed 1,000 pounds.

Shipping

1. Contractor shall ship all deliverables and process development samples to:
   Marshall Space Flight Center
   Central Receiving, Bldg 4631
   Attn: Paul Gradl
   Huntsville, AL 35812

Period of Performance (POP)

- 6 months from authority to proceed (ATP)

Deliverables

1. Witness coupons in the as-built condition from task 10.
2. Build report for witness coupon samples.